

1. Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
KD3-99107-shashank@localhost> select * from orders where cnum = (select cnum from customers where cname = 'Cisneros');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+-----+-----+-----+-----+-----+
2 rows in set (0.001 sec)
```

7. Extract all the orders of Motika.

```
KD3-99107-shashank@localhost> select * from orders where snum = (select snum from salespeople where sname = 'motika');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+-----+-----+-----+-----+-----+
1 row in set (0.002 sec)
```

8. Find all the orders of salespeople servicing customers in London.

```
KD3-99107-shashank@localhost> select * from orders where snum = (select Distinct(snum) from customers where city = 'london');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
3 rows in set (0.003 sec)
```

6. Write a query that selects all orders for amounts greater than any for the customers in london.

```
KD3-99107-shashank@localhost> select * from orders where amt > any (select amt from orders where cnum IN (select cnum from customers where city = 'london'));
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
6 rows in set (0.005 sec)
```

4. Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson)

```
KD3-99107-shashank@localhost> select * from customers where rating >= any (select rating from customers where snum = (select snum from salespeople where sname= 'serres'));
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.001 sec)

11. Select customers who have a greater rating than any other customer in Rome.

```
KD3-99107-shashank@localhost> select * from customers where rating > any(select rating from customers where city = 'rome');
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.001 sec)

12. Select all orders that had amounts that were greater than at least one of the orders from '1990-10-04'.

```
KD3-99107-shashank@localhost> select * from orders where amt > any (select amt from orders where odate = '1990-10-04');
```

onum	amt	odate	cnum	snum
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002
3011	9891.88	1990-10-04	2006	1001

8 rows in set (0.001 sec)

13. Find all orders with amounts smaller than any amount for a customer in San Jose.

```
KD3-99107-shashank@localhost> select * from orders where amt < any(select amt from orders where cnum
IN (select cnum from customers where city = 'san jose'));
+-----+-----+-----+-----+-----+
| onum | amt   | odate  | cnum | snum |
+-----+-----+-----+-----+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 | 75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 | 309.95 | 1990-10-04 | 2004 | 1002 |
+-----+-----+-----+-----+-----+
8 rows in set (0.001 sec)
```

2. Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```
KD3-99107-shashank@localhost> SELECT o.amt, c.cname, c.rating
-> FROM orders o
-> JOIN customers c ON o.cnum = c.cnum
-> WHERE o.amt > (SELECT AVG(amt) FROM orders);
+-----+-----+-----+
| amt   | cname  | rating |
+-----+-----+-----+
| 5160.45 | Liu    | 200    |
| 9891.88 | Clemens | 100    |
| 4723.00 | Clemens | 100    |
+-----+-----+-----+
3 rows in set (0.002 sec)
```

14. Select those customers whose rating are higher than every customer in London.

```
KD3-99107-shashank@localhost> select * from customers where rating > ALL(select rating from customers where city = 'london');
+-----+-----+-----+-----+-----+
| cnum | cname   | city   | rating | snum |
+-----+-----+-----+-----+-----+
| 2002 | Giovanni | Rome   | 200    | 1003 |
| 2003 | Liu      | San Jose | 200    | 1002 |
| 2004 | Grass    | Berlin  | 300    | 1002 |
| 2008 | Cisneros | San Jose | 300    | 1007 |
+-----+-----+-----+-----+-----+
4 rows in set (0.001 sec)
```

5. Write a query that will find all salespeople who have no customers located in their city. (Hint: Choose the correct keyword ANY/ALL).

```
ERROR 1146 (42002): Table 'sales.customer' doesn't exist
[KD3-99107-shashank@localhost> select * from salespeople as s where s.snum not in (select c.snum from customers c where c.city = s.city);
+-----+-----+-----+-----+
| snum | sname   | city   | comm |
+-----+-----+-----+-----+
| 1004 | Motika  | London | 0.11 |
| 1007 | Rifkin  | Barcelona | 0.15 |
| 1003 | Axelrod | New York | 0.10 |
+-----+-----+-----+-----+
3 rows in set (0.001 sec)
```

9. Find names and numbers of all salesperson who have more than one customer.

```
KD3-99107-shashank@localhost> select sname, snum from salespeople where snum IN (select snum from customers group by snum having count(cnum)>1);
+-----+-----+
| sname | snum |
+-----+-----+
| Peel  | 1001 |
| Serres | 1002 |
+-----+-----+
2 rows in set (0.002 sec)
```

10. Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```

Error 1212 (21000): Subquery returning more than 1 row
KD3-99107-shashank@localhost> select snum,sname , city from salespeople where snum in (select snum fr
om customers group by snum having count(cname)>1);
+-----+-----+
| snum | sname | city |
+-----+-----+
| 1001 | Peel | London |
| 1002 | Serres | San Jose |
+-----+-----+
2 rows in set (0.001 sec)

```

3. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```

KD3-99107-shashank@localhost> select snum, sum(amt) from orders group by snum having sum(amt) > max(a
KD3-99107-shashank@localhost> select snum, sum(amt) from orders group by snum having sum(amt) > (sele
ct max(amt) from orders);
+-----+-----+
| snum | sum(amt) |
+-----+-----+
| 1001 | 15382.07 |
+-----+-----+
1 row in set (0.002 sec)

```